SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

   i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right ii. Cost of each move = 1

   iii. Cannot pass through obstacles

   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1.Backtracking Search Solution
Doctors: D1, D2, D3
Shifts: Morning (M), Afternoon (A), Night (N)
Constraints

1. D1 cannot work **Night**.
2. D2 must work **before** D3 (Morning < Afternoon < Night).
3. Only **one doctor per shift**.
4. D3 cannot work **Morning**.
5. Every doctor is assigned **exactly one shift**.

Step 1: Assign D1
Try **Morning** → ✓ Valid

Doctor	**Shift**
D1	Morning

Step 2: Assign D2
Morning is already occupied.
Try **Afternoon** → Temporarily Valid

Doctor	**Shift**
D1	Morning
D2	Afternoon

Step 3: Assign D3
Available shift: **Night**
Constraint Checks:

- D3 cannot work Morning → ✓ Satisfied
- D2 before D3 → Afternoon < Night → ✓ Satisfied
- One doctor per shift → ✓ Satisfied
- All doctors assigned → ✓ Satisfied

Final Valid Schedule

Doctor	**Shift**
D1	**Morning**
D2	**Afternoon**
D3	**Night**

Constraint Verification

- ✓ D1 is not on Night.
- ✓ D2 works before D3.
- ✓ One doctor is assigned to each shift.
- ✓ D3 is not on Morning.
- ✓ Every doctor has exactly one shift.

Final Answer:
D1 → Morning, D2 → Afternoon, D3 → Night.

2. **BFS Search Algorithm (5 × 5 Grid)**

Assume the grid below:

S . . X .
. X . X .
. X . . .
. . X X .
X . . . G

- **S** = Start (0,0)
- **G** = Goal (4,4)
- **X** = Obstacle
- **.** = Free cell
- Movement: Up, Down, Left, Right
- Cost per move = **1**
- BFS finds the shortest path (Manhattan distance is not used by BFS; it is used in heuristic searches such as A*).

Step-by-Step Node Expansion (Queue Updates)

Step	Expanded Node	Queue After Expansion
1	S(0,0)	(0,1), (1,0)
2	(0,1)	(1,0), (0,2)
3	(1,0)	(0,2), (2,0)
4	(0,2)	(2,0), (1,2)
5	(2,0)	(1,2), (3,0)
6	(1,2)	(3,0), (2,2)
7	(3,0)	(2,2), (3,1)
8	(2,2)	(3,1), (2,3)
9	(3,1)	(2,3), (4,1)
10	(2,3)	(4,1), (2,4)
11	(4,1)	(2,4), (4,2)
12	(2,4)	(4,2), (3,4)
13	(4,2)	(3,4), (4,3)
14	(3,4)	(4,3), G(4,4)
15	G(4,4)	Goal Reached

Optimal Path

S → (1,0) → (2,0) → (3,0) → (3,1)
→ (4,1) → (4,2) → (4,3) → G

Path Cost

- Number of moves = **8**
- Cost per move = **1**

Total Cost = 8

Final Answer

- **Optimal Path:** S → (1,0) → (2,0) → (3,0) → (3,1) → (4,1) → (4,2) → (4,3) → G
- **Total Cost: 8, Algorithm Used:** Breadth-First Search (BFS)

3. **3. Autonomous Rescue Robot in a Disaster Building**

(a) Constraint Analysis

Constraint	**Effect on Decision-Making**
Move only Up, Down, Left, Right	Robot considers only four possible moves.
Obstacles cannot be crossed	Robot must avoid blocked cells and find alternate paths.
Limited sensing (adjacent cells only)	Robot has partial knowledge and discovers the environment while moving.
Move cost = 1	Robot prefers shorter paths.
Risky zone cost = +2	Robot avoids risky areas unless necessary to reduce total cost.
Dynamic knowledge update	Robot updates its map whenever new cells are observed.

(b) Solution Approach (Online Uniform Cost Search)

1. Start at **S** with an empty map.
2. Sense adjacent cells and identify free, blocked, and risky cells.
3. Calculate the cumulative path cost:
   - Normal cell = **1**
   - Risky cell = **3 (1 + 2)**
4. Expand the node with the **lowest total cost**.
5. Update the map whenever new information is discovered.
6. Repeat until the robot reaches **Goal (G)**.

(c) AI Concepts Applied

1. Intelligent Agent

- The rescue robot acts as an intelligent agent.
- It senses the environment, processes information, and performs actions to reach the survivor.

2. Rationality

- The robot always selects the action with the **minimum total cost** while avoiding obstacles and risky zones.
- It makes the best decision based on the information currently available.

3. Environment Type

- **Partially Observable:** Only adjacent cells are visible.
- **Dynamic:** New information is discovered during movement.
- **Sequential:** Each move affects future decisions.
- **Discrete:** The environment consists of grid cells.

(d) Justification of the Approach

Online Uniform Cost Search is the most suitable because:

- It works well in **partially observable environments**.
- It updates the search as new information becomes available.

- It always chooses the **least-cost path**.
- It avoids risky zones whenever a safer, lower-cost route exists.
- It guarantees an optimal solution when all movement costs are non-negative.

Final Answer

- **Search Strategy:** Online Uniform Cost Search (UCS)
- **Advantages:** Handles unknown environments, updates knowledge dynamically, minimizes total cost, and safely navigates around obstacles and risky zones.
- **AI Concepts Used:** Intelligent Agent, Rationality, Partially Observable Environment, Dynamic Knowledge Updating.